

ECO303

Midterm Exam II

Instructions:

This exam has two questions. The total number of points for the exam is 110. I will grade it over 100 points. **Write your answers for question 1 only on blue sheets of paper and write your answers for question 2 only on pink sheets of paper. Please write your name and ID on every sheet that you use.** This exam is closed notes/books. Calculators are allowed but other electronic devices are **not**. For questions with a numerical answer, give a numerical answer. Good luck!

Do not turn this page unless instructed to!

- I. **Labor and time.** Mr. Pusheen lives for two periods, say when he is young and when he is old. Let $t = 1$ denote the period when he is young and $t = 2$ denote the period when he is old. In period $t = 1$, Mr. Pusheen is endowed with total hours $\bar{R} = 175,000$ which he can use between working and relaxing. Working is the only source of income of Mr. Pusheen when young. Mr. Pusheen receives an hourly wage rate of w period $t = 1$. On the other hand, Mr. Pusheen expects to retire when he is old, in period $t = 2$, which means that he will not work in this period, in fact, Pusheen only derives utility from consumption in period 2. Any income available for Mr. Pusheen in period $t = 2$ comes from savings in period 1. Mr. Pusheen can save part of his labor income earnings in period $t = 1$ at an interest rate of τ . Let c_1 and c_2 denote consumption in periods 1 and 2 respectively; R_1 and L_1 denote relaxation time and working time in period 1; finally s_1 denotes savings in period 1. Mr. Pusheen derives utility from consumption and relaxation as follows

$$U(c_1, c_2, R_1) = \log(c_1) + \log(R_1) + \delta \log(c_2).$$

Assume that price of consumption in both periods is equal to 1.

- 1) (6 points) State Mr. Pusheen's period 1 budget constraint.
 - 2) (6 points) State Mr. Pusheen's period 2 budget constraint.
 - 3) (10 points) Using the previous two items, use s_1 to consolidate the per-period budget constraints into the *intertemporal budget constraint*. Hint: The only variables that should appear in the intertemporal budget constraint are c_1, c_2 and R_1 .
 - 4) (8 points) State Mr. Pusheen's utility maximization problem.
 - 5) (10 points) Solve for the optimal levels of consumption and the optimal labor supply of Mr. Pusheen.
 - 6) (10 points) How do the optimal levels of consumption and the labor supply change with respect to changes in w ? Provide a formal answer and economic intuition.
- II. **Profit maximization.** Pip owns a business that produces widgets y . In the short-run, producing widgets y is a labor-intensive operation. It is produced using labor with the production function $y = l^{\frac{1}{3}}$. Pip pays his workers a wage rate of w . Let the price of widgets be p . Pip maximizes his profits as follows:

$$\max_{l \geq 0} pl^{\frac{1}{3}} - wl,$$

- 1) (6 points) Does the production function exhibits constant, increasing or decreasing returns to scale?
- 2) (6 points) Obtain the first-order condition from Pip's profit maximization problem and interpret it.
- 3) (8 points) Use item (2) to solve for Pip's optimal labor input demand and the optimal supply of widgets.

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- 4) (10 points) Suppose that the labor market is risky so that w fluctuates from month-to-month, and Pip needs to decide in advance how to produce and how many workers to hire before the wage rate is known. Assume that the wage rate w is 10 with probability ψ , and 5 with probability $1 - \psi$. Pip should maximize *expected profits*. State the maximization problem and explain why you are setting up the problem the way you are.
- 5) (6 points) Derive Pip's first order condition for the problem in item (4) and interpret it.
- 6) (8 points) Use item (5) to solve for Pip's optimal level of labor demand and optimal supply of widgets.
- 7) (8 points) How do the optimal levels of l and y change with respect to the probability ψ ? Provide economic intuition for your answer.
- 8) (8 points) Suppose instead that Pip knew that the wage rate was going to be for sure $\bar{w} = \psi 10 + (1 - \psi) 5$, which is the expected value of the uncertain wage rate. Using your solution for point 3, compute the quantity of labor hired by Pip in this case, and compare it to the quantity hired when the wage is uncertain (the one from point 6). How do the two quantities compare? Discuss the reason for this result.
- 9) (10 points) Think about the situation described in items (4)–(8). Would Pip be willing to pay to write a contract that fixes the wage rate at $\bar{w}$? Provide an argument for your answer.